

BEFAST AI — Conclusion

Rule-based screening for the BE-FAST stroke warning signs — evaluation methodology, limitations and future work

1. Summary of the work

BEFAST AI is a browser-based prototype that guides a user through the six BE-FAST stroke warning signs. Balance, Eyes, Face and Arms are measured from MediaPipe pose- and face-landmark streams; Speech is assessed by comparing a Web Speech API transcript against a fixed target phrase; Time is not sensed but recorded from an onset question and carried as a running clock into a responder handoff summary. Scoring is rule-based rather than learned: fifteen thresholds in a single module map continuous measures onto four per-sign states, and an observer questionnaire serves as both fallback and override. The system is biased toward false alarms — one uncertain sign opens the emergency route.

Camera analysis is genuinely local; frames are never transmitted. The claim of wholly on-device processing shown in the interface is nevertheless inaccurate for Speech, because the Web Speech API transcribes through a vendor cloud service in every browser that implements it. Model weights and the WASM runtime are also fetched from public CDNs on first use.

2. Accuracy has not been measured

The accuracy of this prototype is unknown and no claim is made about it. The thresholds were chosen by inspection and the source declares them unvalidated; there is no labelled dataset, no reference standard and no automated test suite. The interface therefore reports categorical urgency tiers rather than a probability, and tags every displayed figure a prototype estimate. Those fifteen thresholds are also not the whole decision surface: feature weights, the ten-second hold, the 2.2-second target exposure and the eight-second recording window are undocumented constants inside the modules.

3. Construct validity must be established first

Review of the feature extraction found defects that would bias any clinical study and should be corrected before recruitment. Arm drift is divided by an arm length recomputed on the final frame, so as the arm falls the divisor shrinks and the normalised drift inflates; only the first and last frames contribute, making a drift that recovers within the hold invisible, and the comparison is signed, so upward drift can never trip a threshold. Postural sway is a standard deviation normalised by the last frame's shoulder width, diluting a single large lurch. The eyelid and gaze figure is whatever the final animation frame produced, with no accumulation, so an ordinary blink can determine the result. Transcript accuracy is order-insensitive bag-of-words matching, so a scrambled utterance of the correct words scores 1.0, and pauses are counted only when voice resumes, so trailing silence is never counted. The scored facial asymmetry index omits the mouth-frown term that the live on-screen meter includes.

4. How accuracy should be measured

Reference standard. Ground truth should be the discharge diagnosis established by neuroimaging and neurologist assessment; a lower-cost first study may substitute concurrent NIHSS or FAST-ED scoring by a certified examiner. Labels are required per sign, not only per patient.

Study design. A prospective paired-comparison study in an emergency department, with the application and the reference standard blinded to each other. Recruitment must include stroke mimics — Bell's palsy, migraine, seizure, hypoglycaemia — and healthy controls; specificity estimated from a stroke-only cohort is meaningless.

Primary metrics. Sensitivity, specificity, PPV and NPV with 95% confidence intervals, per sign and for the screen as a whole, from a 2x2 contingency table. For an instrument whose output is "call emergency services", sensitivity is the quantity to maximise: a false negative is the harmful error, whereas a false positive costs one avoidable ambulance call.

Threshold selection. The continuous features should be swept across ROC curves, with AUC reported and operating points fixed at a pre-specified minimum sensitivity rather than by eye. Three output tiers imply two thresholds per sign, so an explicit misclassification-cost criterion is required.

Agreement and reliability. Cohen's kappa against clinician judgement and between observers answering the questionnaire; Bland-Altman analysis of the continuous measures against manual annotation; test-retest repeatability; and inter-device variance across handsets, lighting and camera distance, since landmark quality depends on all three.

Power and subgroups. The cohort should be sized for a target confidence interval on sensitivity, with subgroup analysis by skin tone, age, eyewear, facial hair, seated posture and language.

5. Software prerequisites

Each measurement is formatted into a display string and the underlying number discarded; at most twenty archived sessions are held in browser storage, and there is no consented export path. Evaluation therefore requires numeric feature logging, an export mechanism, an offline replay harness that re-runs the scoring functions over recorded media so thresholds can be re-swept without re-recruiting, and unit tests around the decision logic.

6. Limitations

A single camera yields 2D landmarks that only approximate 3D motion, degrading with poor lighting, oblique angles and distance. No service worker is present, so despite an installable manifest the application cannot run offline. Speech recognition is absent in Firefox and inconsistent on iOS Safari, in which case the sign falls back to the questionnaire. The target phrase is English with the recogniser fixed to en-US, and would score healthy Thai speakers as impaired. BE-FAST does not capture every posterior-circulation presentation. This is not a diagnostic device.

7. Conclusion

The prototype demonstrates that a complete BE-FAST pathway — five sensed signs plus onset timing, an auditable rule-based scorer and a fail-safe bias toward escalation — can run in a browser with no application server. It remains an engineering demonstration rather than a validated instrument: the defects in Section 3 must be corrected and the study in Section 4 conducted before any statement about its accuracy is possible.